TASK - 2

PART C - Deriving Attention Gradients

① $S = \frac{QK^T}{\sqrt{d_k}}$ (Attention scores)

② $P = \text{softmax}(S)$ (softmax probabilities)

③ $A = PV$ (final attention output)

where,

$Q \in \mathbb{R}^{T \times d_k}$
$K \in \mathbb{R}^{T \times d_k}$
$V \in \mathbb{R}^{T \times d_v}$ } input embeddings
$P \in \mathbb{R}^{T \times T}$
$A \in \mathbb{R}^{T \times d_v}$

T = no. of tokens
d_k

17) Derive $\frac{\partial A}{\partial V}$

→ $A = PV$
$P \to$ constant ⇒ PV is a matrix multiplication which is linear in V.

$$A_{ij} = \sum_k P_{ik} V_{kj}$$

We will differentiate w.r.t. V_{ab} (some a row b column element of V)

$$\frac{\partial A_{ij}}{\partial V_{ab}} = \frac{\partial}{\partial V_{ab}} \left(\sum_k P_{ik} V_{kj} \right)$$

→ Derivative exists only when $j = b$, $k = a$

$k = a$

$$\Rightarrow \frac{\partial A_{ij}}{\partial V_{ab}} = \frac{\partial (P_{ia} V_{aj})}{\partial V_{ab}}$$

for $j = b$, derivative $= P_{ia}$
else $= 0$

$$\Rightarrow \frac{\partial A_{ij}}{\partial V_{ab}} = \begin{cases} P_{ia}, & j = b \\ 0, & \text{otherwise} \end{cases}$$

$$\Rightarrow \boxed{\frac{\partial A_{ij}}{\partial V_{ab}} = P_{ia} \underbrace{\delta_{jb}}}$$

Kronecker Delta $f^n = \begin{cases} 1 & j = b \\ 0 & j \neq b \end{cases}$

18) Show that Jacobian $\frac{\partial p}{\partial s}$ has entries : $\frac{\partial p_i}{\partial s_j} = p_i(\delta_{ij} - p_j)$

→ To do : derive $\frac{\partial p_i}{\partial s_j}$

$$p_i = \frac{e^{s_i}}{\sum_k e^{s_k}} = \frac{e^{s_i}}{Z}$$

$\Rightarrow \frac{\partial p_i}{\partial s_j}$:

Case ① : $i=j$

$$\Rightarrow \frac{\partial p_i}{\partial s_i} = \frac{Z \cdot \frac{\partial}{\partial s_i}(e^{s_i}) - e^{s_i} \frac{\partial}{\partial s_i} \cdot \sum_k e^{s_k}}{Z^2}$$

$$= \frac{e^{s_i} Z - e^{s_i} e^{s_i}}{Z^2}$$

$$= \frac{e^{s_i}}{Z}\left(1 - \frac{e^{s_i}}{Z}\right)$$

$$\Rightarrow \boxed{\frac{\partial p_i}{\partial s_i} = p_i(1-p_i)}$$

Case ② : $i \neq j$

$$\Rightarrow \frac{\partial p_i}{\partial s_j} = \frac{Z \cdot \frac{\partial}{\partial s_j}(e^{s_i}) - e^{s_i} \frac{\partial}{\partial s_j} \cdot \sum_k e^{s_k}}{Z^2}$$

$$= \frac{0 \cdot Z - e^{s_i} e^{s_j}}{Z^2}$$

$$= \frac{e^{s_i}}{Z}\left(0 - \frac{e^{s_j}}{Z}\right)$$

$$\Rightarrow \boxed{\frac{\partial p_i}{\partial s_j} = p_i(0-p_j)}$$

Combining both, its clearly seen : $\boxed{\frac{\partial p_i}{\partial s_j} = p_i(\delta_{ij} - p_j)}$

19) Derive $\frac{\partial A}{\partial Q} = \left(\frac{\partial A}{\partial P}\right)\left(\frac{\partial P}{\partial S}\right)\left(\frac{\partial S}{\partial Q}\right)$

$$\rightarrow \boxed{\frac{\partial A}{\partial P} = \frac{\partial A_{ij}}{\partial P_{ab}} = \delta_{ia} V_{bj}}$$

$$\boxed{\frac{\partial P_{ab}}{\partial S_{ac}} = P_{ab}(\delta_{bc} - P_{ac})}$$

$$S = \frac{QK^T}{\sqrt{d_k}} \quad : \quad S_{ab} = \frac{1}{\sqrt{d_k}} \sum_m Q_{am} K_{bm}$$

$$\Rightarrow \boxed{\frac{\partial S_{ab}}{\partial Q_{pq}} = \frac{1}{\sqrt{d_k}} \delta_{ap} K_{bq}}$$

$$\Rightarrow \frac{\partial A_{ij}}{\partial Q_{pq}} = \sum_{a,b,c} \frac{\partial A_{ij}}{\partial P_{ab}} \cdot \frac{\partial P_{ab}}{\partial S_{ac}} \cdot \frac{\partial S_{ac}}{\partial Q_{pq}}$$

$$= \sum_{a,b,c} (\delta_{ia} V_{bj}) P_{ab}(\delta_{bc} - P_{ac}) \frac{1}{\sqrt{d_k}} \delta_{ap} K_{cq}$$

$\delta_{ia} \circ \delta_{ap} = 1$ only when $i = a = p$

$$\Rightarrow \boxed{\frac{\partial A_{ij}}{\partial Q_{pq}} = \frac{\delta_{ip}}{\sqrt{d_k}} \sum_{b,c} V_{bj} P_{ib} (\delta_{bc} - P_{ic}) K_{cq}}$$

20) Interpretation

As the dimension d_k increases, the dot products in QK^T become very large. When these large values are passed through softmax, the output prob. become one sided, i.e, some values very close to 1 rest very close to 0.

$\frac{\partial P_i}{\partial s_j} = P_i(\delta_{ij} - P_j) \rightarrow$ here gradient will become very small. which will cause unstability in training.

So scaling by $\sqrt{d_k}$ prevents this softmax saturation.